

Please ensure this tracking sheet is completed as a group, as it will be used to monitor your individual participation within the team.

Team Members	Role/ Responsibility	Email				
Belyse Intwaza	Team Lead	b.intwaza@alustudent.com				
Saad Byiringiro	Team Member	s.byiringir@alustudent.com				
Task Allocation and Tracker						
	Task Allocated	Assigned Member	Deadline	Completion Status (Not Started/In Progress/Completed/Did not participate)	Reviewed by Team? (Yes/No)	Comments
	Find an African dataset (NISR Rwanda Seasonal Agricultural Survey 2024 Season B), upload raw data and set up GitHub repo	Belyse Intwaza	30/09/2026	Completed	Yes	Raw CSV has missing values and text columns; metadata added; repo link added to notebook
	Data preparation: explore columns, handle missing values, encode text columns, save processed CSV	Belyse Intwaza	30/09/2026	Completed	Yes	01_data_preparation_and_pca.ipynb; 17,260 rows x 20 features, no missing values
	Task 1: PCA from scratch (centering, covariance matrix, eigenvalues/eigenvectors, sorting, projection)	Belyse Intwaza	30/09/2026	Completed	Yes	01_data_preparation_and_pca.ipynb; numpy + matplotlib only, checked against np.cov
	Before/after PCA visualisations and interpretation of PC1 and PC2	Belyse Intwaza	30/09/2026	Completed	Yes	Covariance heatmap, before_pca.png and after_pca.png in figures/
	Task 2: explained variance, cumulative variance, dynamic selection of k (scree and cumulative plots, information lost per feature)	Saad Byiringiro	30/09/2026	Completed	Yes	02_component_selection_and_benchmarking.ipynb; 90% threshold gives k = 15 (92.1% of variance kept)
	Task 2: written explanation (why k, tradeoff, information lost)	Saad Byiringiro	30/09/2026	Completed	Yes	Markdown cell in 02_component_selection_and_benchmarking.ipynb
	Task 3: optimized chunked PCA, benchmark of time and memory, 50-million-row streaming test	Saad Byiringiro	30/09/2026	Completed	Yes	About 3x faster and 3.5x less memory at 3M rows; same results as the original
	Task 3: written explanation of the improvements, and README update	Saad Byiringiro	30/09/2026	Completed	Yes	Markdown cell in 02_component_selection_and_benchmarking.ipynb; README lists both notebooks
	Final notebook review, combined PDF and submission	Belyse Intwaza, Saad Byiringiro	30/09/2026	Completed	Yes	Both check before submitting

Meeting Attendance Log						
Meeting Date	Agenda	Facilitator	Attendees	Key Discussion Points	Next Steps	
25/09/2026	Introductions, go through the assignment brief together and allocate tasks		Belyse Intwaza, Saad Byiringiro	Read the brief together and split the work: Belyse on dataset, data preparation and Task 1; Saad on Task 2, Task 3 and interpretation	Belyse to find an African dataset for the assignment	
26/09/2026	Check the dataset Belyse found		Belyse Intwaza, Saad Byiringiro	Reviewed the NISR Rwanda Seasonal Agricultural Survey 2024 Season B data we will work with	Belyse: data preparation and Task 1; Saad: Task 2 and Task 3	
30/09/2026	Final review and submission		Belyse Intwaza, Saad Byiringiro	Final review of the notebooks, explanations and task sheet before submitting	Submit the assignment	